Evaluating Digital Training and an AI Coach for ECD

The purpose of this document is to develop research plans that we can layer on top of implementation work that Dimagi is doing with its CommCare Connect Program for Early Childhood Development (CCC-ECD).

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Description of the CCC-ECD program

In CCC-ECD the following steps happen:

1. Onboarding: Frontline workers (FLWs) are onboarded to CommCare Connect by an Locally-led Org (LLO) on their own smartphone in a 1-day training. They fill out a user survey with demographic information, how comfortable they are with digital technology, etc. They are added to a Whatsapp group chat with other FLWs they are onboarded with.
2. Job offering: After a few days, FLWs are notified on their smartphone that they have a CommCare Connect job opportunity to learn an ECD intervention. Through the app they are told the scope of the opportunity in terms of how much they will be paid per client, for how many clients, within a given time window (a few months).
3. Digital training: FLWs who opt in then take an offline, mobile training course at their own pace. They need to pass a knowledge test to continue. If they fail, they can retake the course and the test as many times as they want.
4. Unsupervised practice: FLWs who pass the knowledge test, are then paid to practice the intervention twice, with digital support from a CommCare Connect Deliver app --e.g., on their family or friends.
5. Unsupervised delivery: FLWs are then paid to deliver the intervention in their communities, using the same Delivery app, which records timestamps and GPS coordinates.
6. AI-coaching: The FLWs are also required to engage with an AI coach over Whatsapp in order to continue with the CommCare Connect opportunity. See below for what occurs during those coaching sessions. The schedule can vary from a daily to weekly AI coaching session.

   Note: steps 2-6 happen without any communication with the LLO. They are entirely automated by the CommCare Connect platform.
7. Supervised observation visit #1: After the FLWs begin to deliver the intervention in their communities, the LLO arranges for a field supervisor to observe each FLW during a household session - aiming for this to occur after the FLW has completed about five unsupervised sessions. The field supervisor observes the FLW, and fills out a score card on a supervisor app that rates the FLW on their performance. The supervisor also provides feedback to the FLW.
8. Supervised observation visit #2: The LLO arranges a second visit to occur after the FLW has more visits, and repeats the same scorecard and feedback.

AI Coach for CCC-ECD

We have experimented with AI-based chatbots built on Dimagi's Open Chat Studio platform for building AI chatbots with Large Language Models, such as GPT-4.

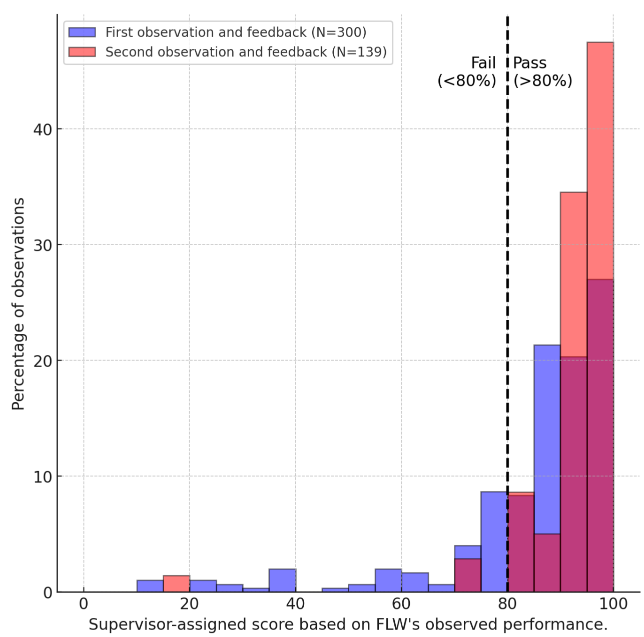
As part of the CommCare Connect program, FLWs are required to interact with our AI Coach over WhatsApp. The AI Coach leads the session and determines when the FLWs have engaged sufficiently. (Currently the AI coach then gives the FLW a code which they can enter into their CommCare Connect app, though we'll automate that link.) The AI Coach speaks in an appropriate local language. We are experimenting with different coaching behaviors, including:

- debriefing with the FLWs about their recent visits
- quizzing the FLW about ECD knowledge
- role-playing with the FLWs to teach them techniques such as motivational interviewing
- answering questions the FLWs have about ECD instruction

Progress in 2024 on CCC-ECD and Targets for 2025

In 2024, we conducted pilots of CCC-ECD in Malawi, Mozambique, and Nigeria. These pilots used our current CCC-ECD program, which involves the FLW conducting three ECD training visits delivered at least 1 week apart. To date, over 39,000 CCC-ECD visits have been conducted by over 450 FLWs across the three countries. The pilots indicate strong demand for the program, with 71% of FLWs successfully progressing through the digital learning phase to service delivery. Initial quality assessments showed that 78% of FLWs scored above 80% during their first observed visits, demonstrating their capability to deliver the intervention effectively. Moreover, follow-up observations revealed further improvement in scores, with average performance increasing from 85% in the first observation to 91% in the second.

We are targeting to scale up CCC-ECD to around 100,000 more visits by Q1 2026.



Research Question #1: Can FLWs Learn to Deliver an ECD intervention without any Human-led Training?

This research question relates to assessing our success on the primary objective we focused on with CCC-ECD in 2024-- i.e., could we digitally train FLWs to conduct the ECD intervention?

The problem we're solving is different from other digital training work for FLWs that we are aware of. To our knowledge, most digital training of FLWs falls into two categories. The first category involves conducting hybrid training with in-person training interspersed with remote, online digital learning. This helps standardize the instruction and reduce training costs. An early example is the Novartis Foundation's IMCI Computerized Adaptation and Training Tool (ICATT). A more recent example is work supported by Last Mile Health in Ethiopia, in which they reduced costs from $605 to $372 per trainee by shifting from a training that required 10 in-person sessions to one that required 4 in-person sessions and 5 digital sessions. The second category of training is online, opt-in courses that FLWs can take to build knowledge (vs. to enable FLWs to perform some new intervention). Some of these courses include a knowledge test at the end. The focus of these courses is knowledge refreshing or knowledge improvement, but not to teach and validate a new skill.

In contrast, we want to know if FLWs can learn to deliver a new intervention through digital-only training and unsupervised practice that is validated by a supervised, practical test.

Our initial results are encouraging. We are interested in partnering with researchers to analyze our existing data and/or conduct additional research on top of our planned 2025 CCC-ECD scale up.

Research Question #2: Can an AI Coaching Chatbot Improve FLW Delivery of an ECD Intervention and Identify Low-performers?

Our 2024 pilots included some preliminary efforts to introduce an AI coach. We are planning to do much more with the AI Coach in 2025. Our primary hypothesis is:

AI-Coached FLWs will implement CCC-ECD better than non AI-Coached FLWs.

Our secondary hypotheses are:

We can use the AI-coach interactions to identify low performers

FLWs will like and benefit from the AI coach

Our plan is to randomize FLWs to different variations of the AI Coach in 2025, and use the supervisory observed visits (steps 7 and 8) as the primary outcome for our primary hypothesis. We can also assess the impact of the AI coach on other aspects of implementing CCC-ECD well, including time spent in visits, and how many ECD visits they complete.

The data we collect will also serve as training and test data for our secondary hypothesis about predicting behavior. Dimagi and partners can run a Machine Learning process to predict the practical test results from the AI transcripts that occur prior to the practical test.

There are several things we can vary in the AI coach. We can vary the frequency and duration of the sessions. We can vary the focus of the sessions, including debriefing on existing visits, quizzing and retraining on ECD content, or role-playing complex cases. We can randomize workers to a control chatbot that does minimal ECD training to isolate the impact of the other functions of the AI coach.

Research Question #3: Can a CCC-ECD program improve caregiver behavior and child development outcomes?

Dimagi is seeking funding to expand our initial work on CCC-ECD. We are interested in expanding the current three visit ECD program to one that has 8-10 visits, likely spread out across the first year of an infant's life. We would like to evaluate the impact of this CCC-ECD program on caregiver behavior and potentially on early childhood development. Please contact us if you would like to partner on this effort.